CS5493 Topics in Autonomous Driving

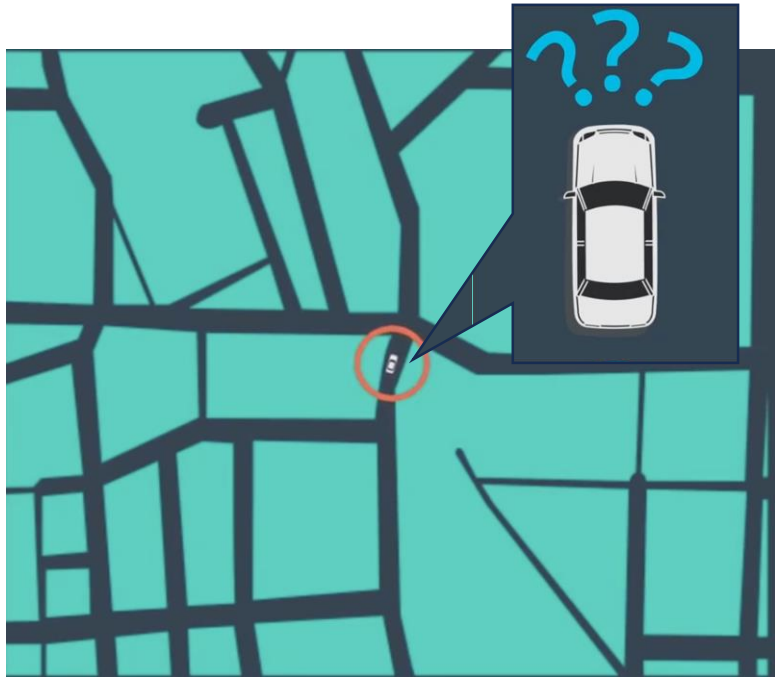
香港城市大學
City University of Hong Kong
Innovating into the Future

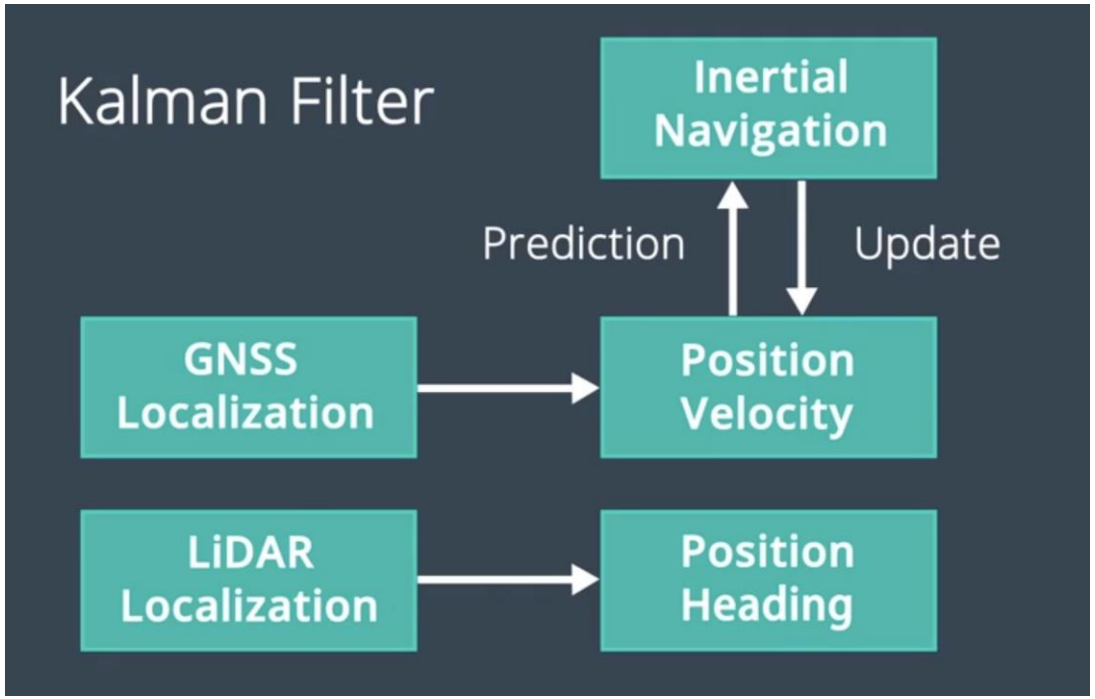
Tutorial 8:
Localization & Planning in Apollo

Localization in Autonomous Driving

Knowing the current position of the vehicle on the HD Map is essential for the AD systems to perform tasks such as path planning.

Modern autonomous driving systems, including Apollo, adopt multiple sensors to achieve real-time and highly accurate localization.

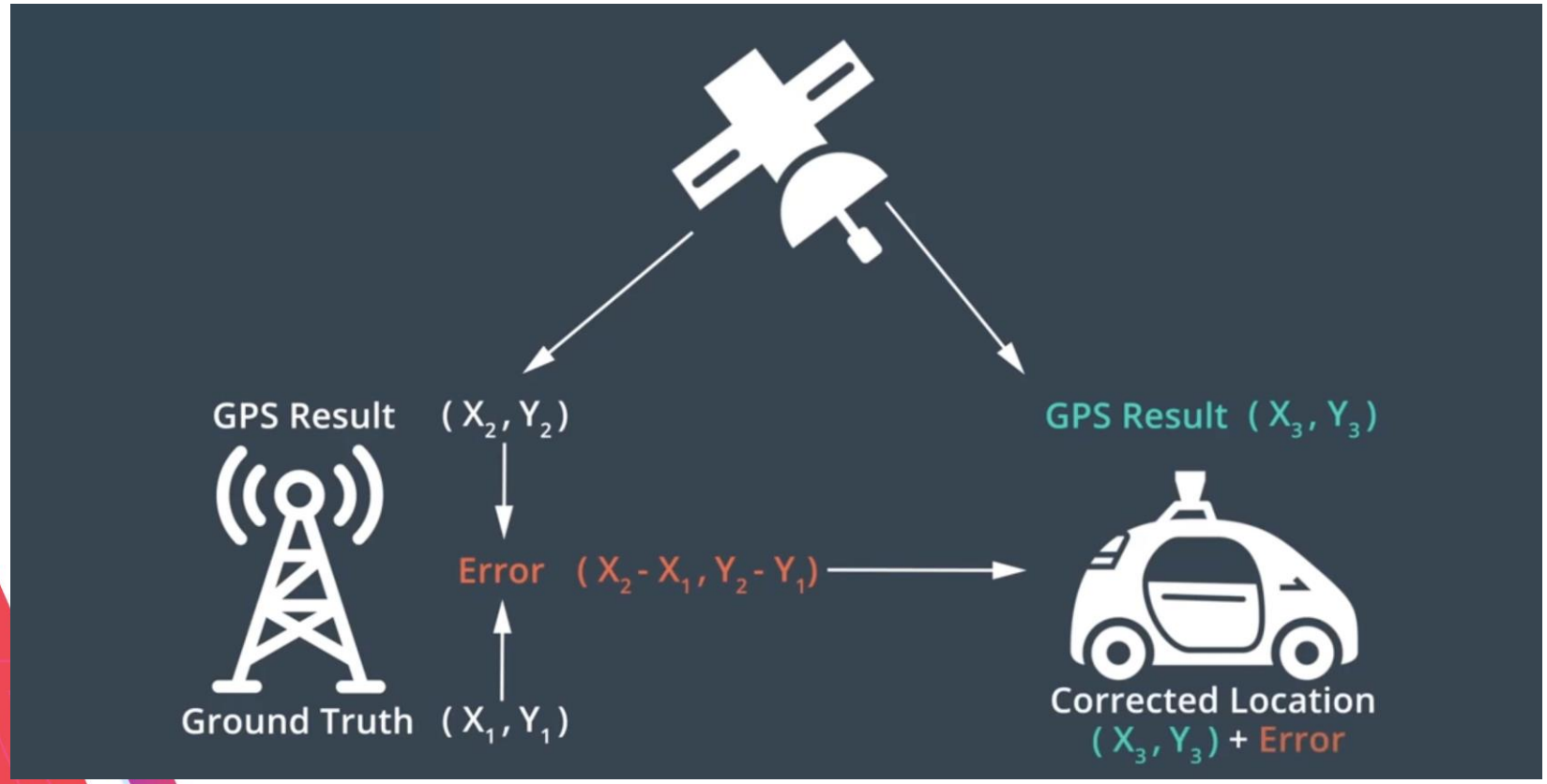




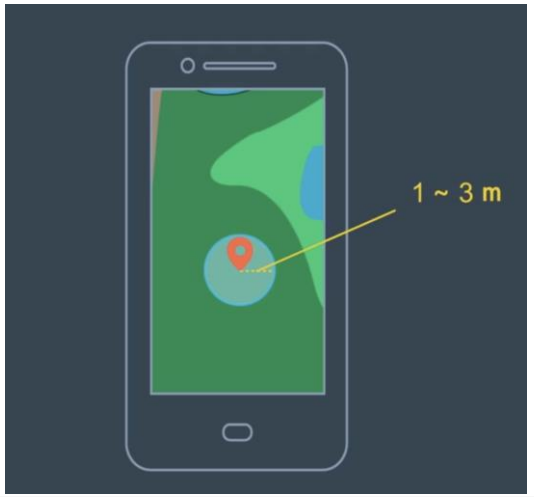
Localization in Apollo

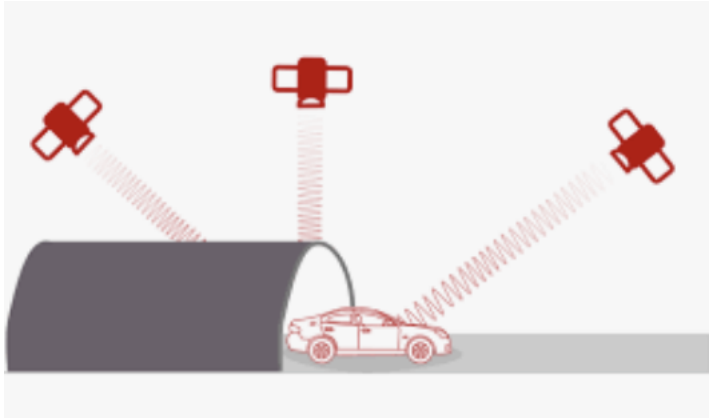
Localization in Apollo: GNSSRTK

GNSSRTK is very straightforward but also has some inevitable flaws.



GNSSRTK Localization Overview



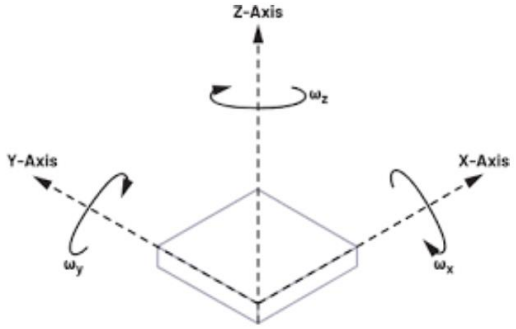


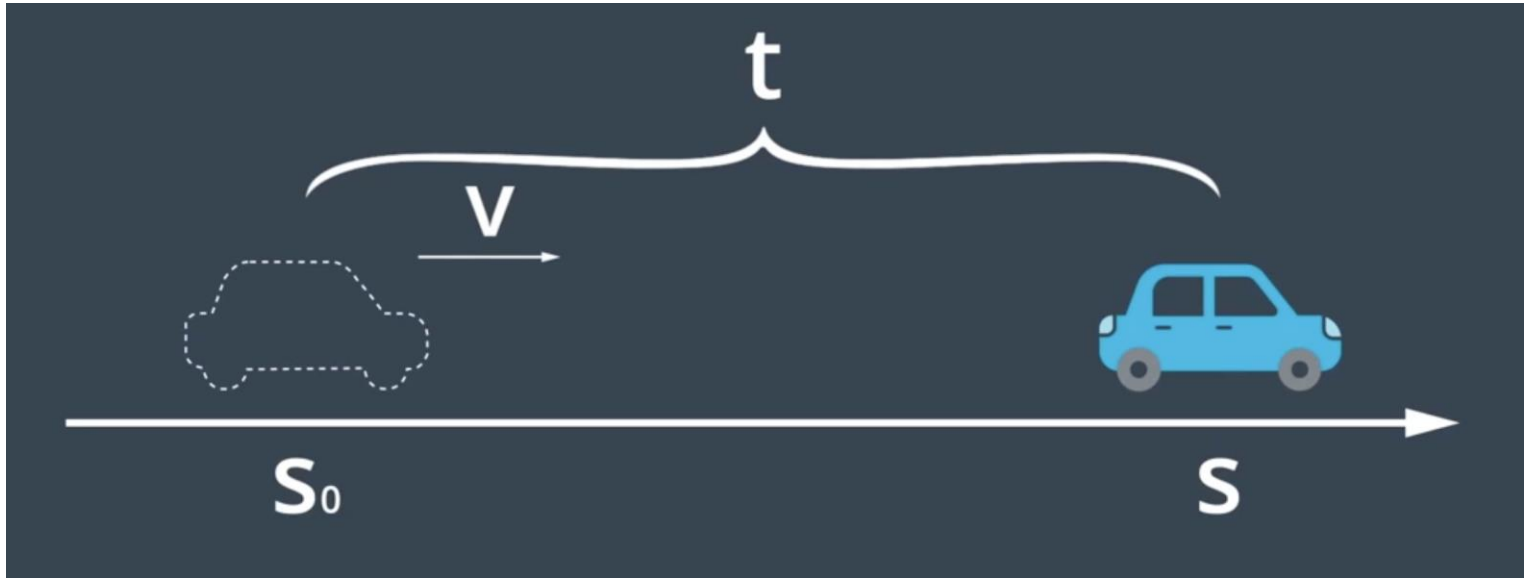
Localization in Apollo: Inertial Localization

Inertial localization is also a very common approach with inertial sensors such as 3D gyroscope and IMU.

3D Gyroscope

IMU

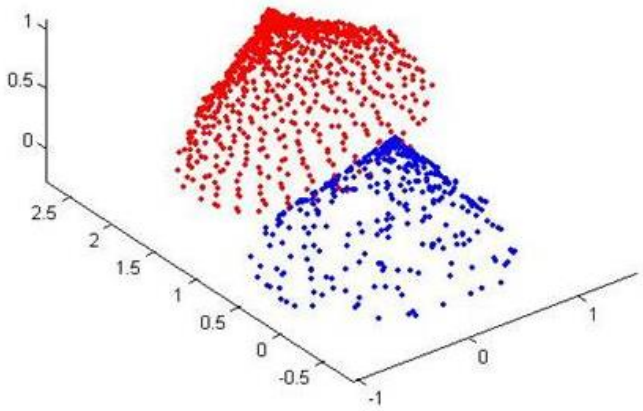




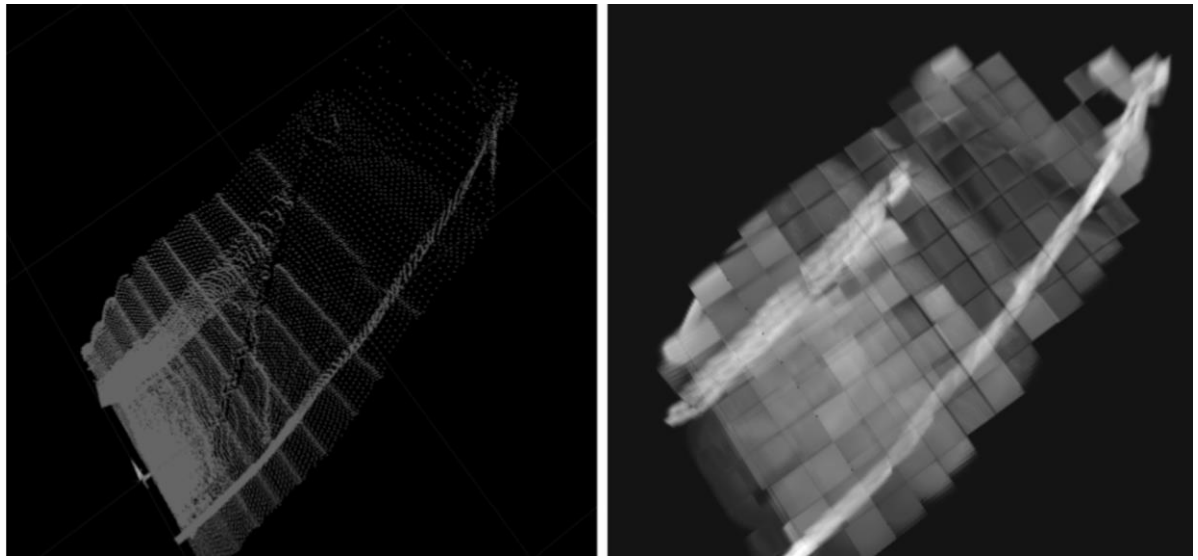
Localization in Apollo: LiDAR Localization

LiDAR Localization adopted in Apollo to provide precise position of the vehicle on the HD Map.

- **ICP (Iterative Closest Points)**: Uses point-to-point / point-to-plane correspondence, simple but depends strongly on initial pose and is slower for large point clouds.
- **NDT (Normal Distributions Transform)**: Uses probability distribution-based matching, less sensitive to noise, faster in large scenes.



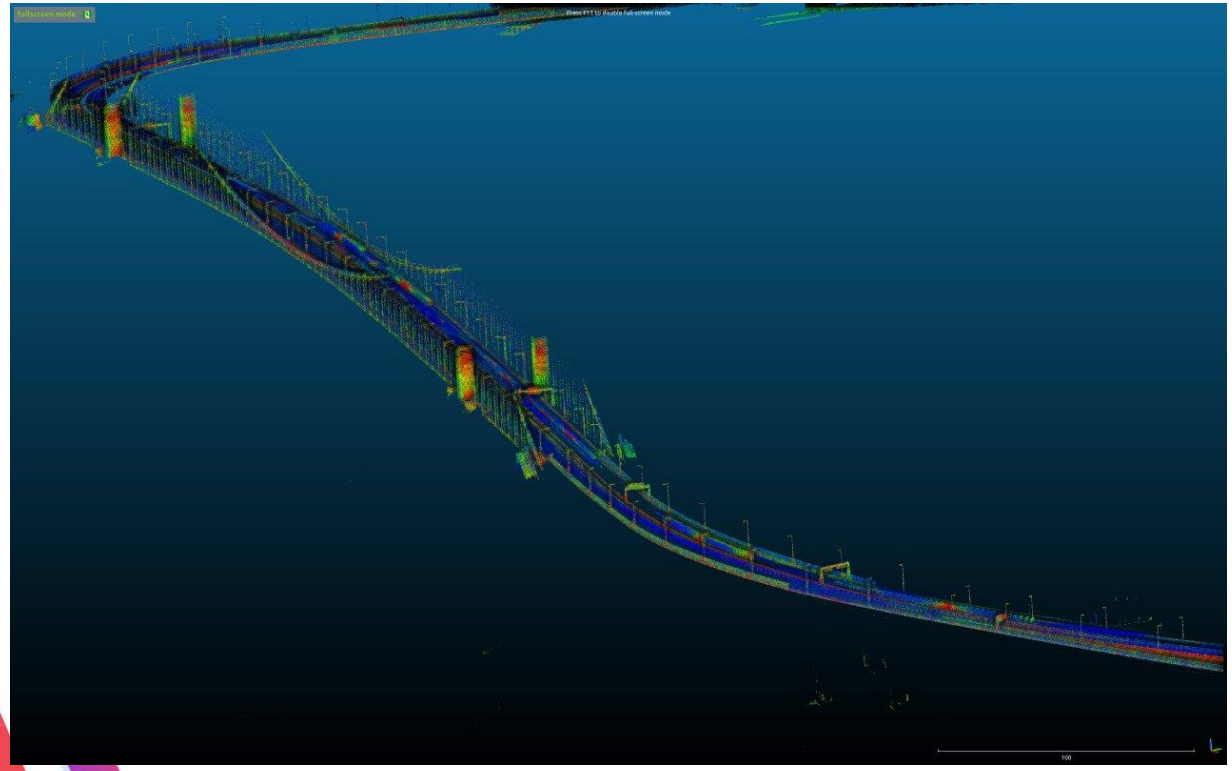
ICP (Iterative Closest Points)

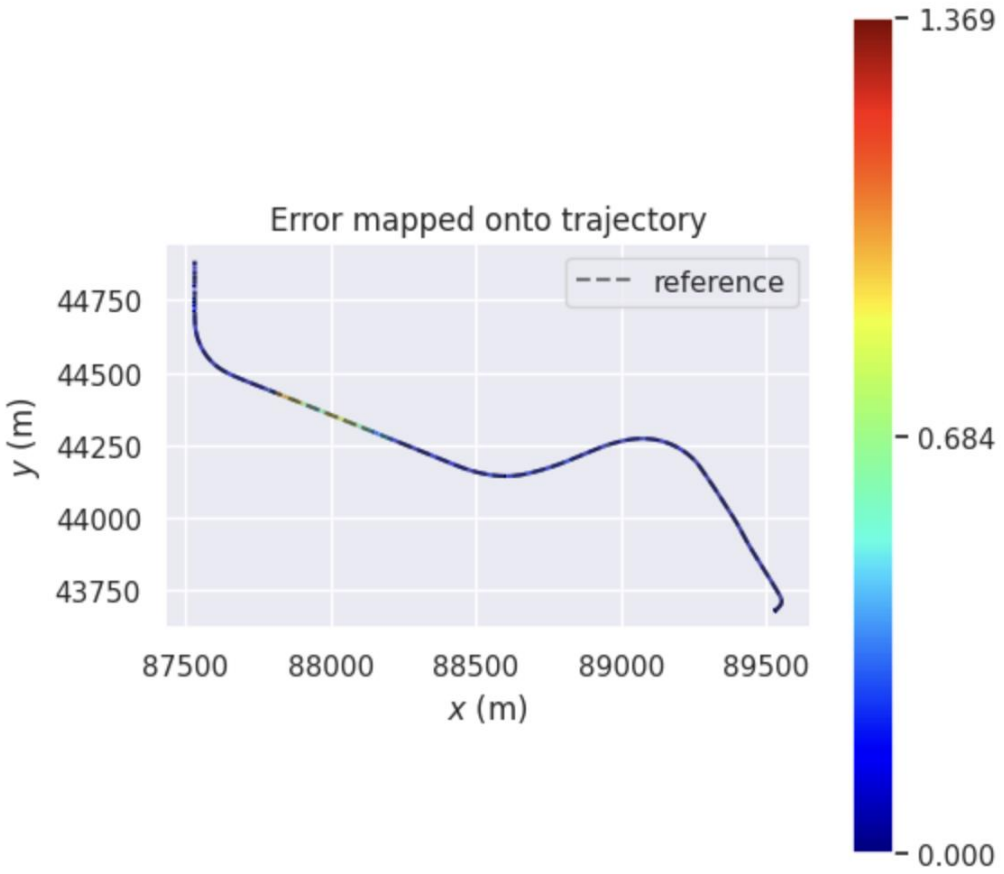


NDT: Process the raw points to probability distribution

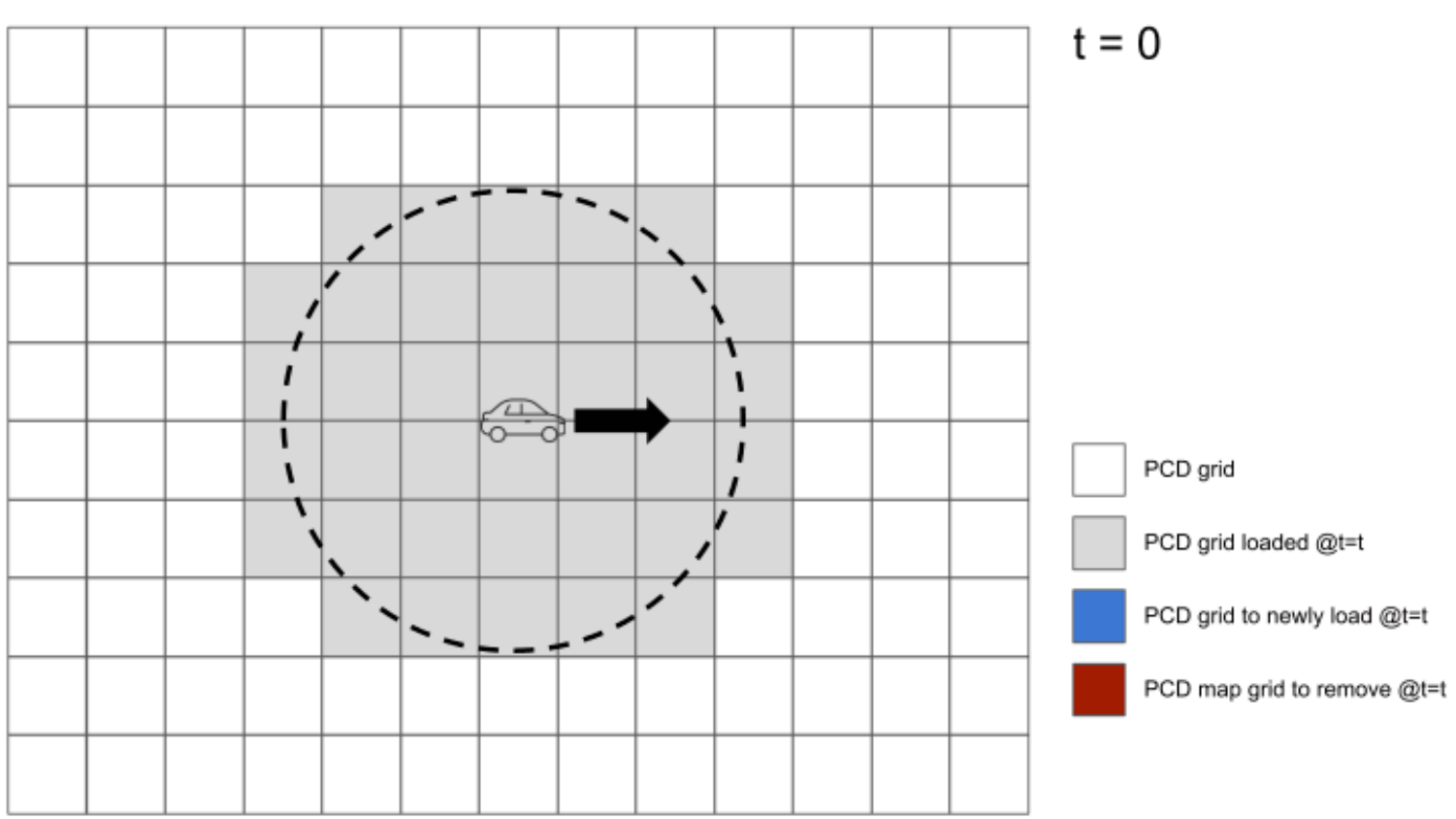
Localization in Apollo: LiDAR Localization

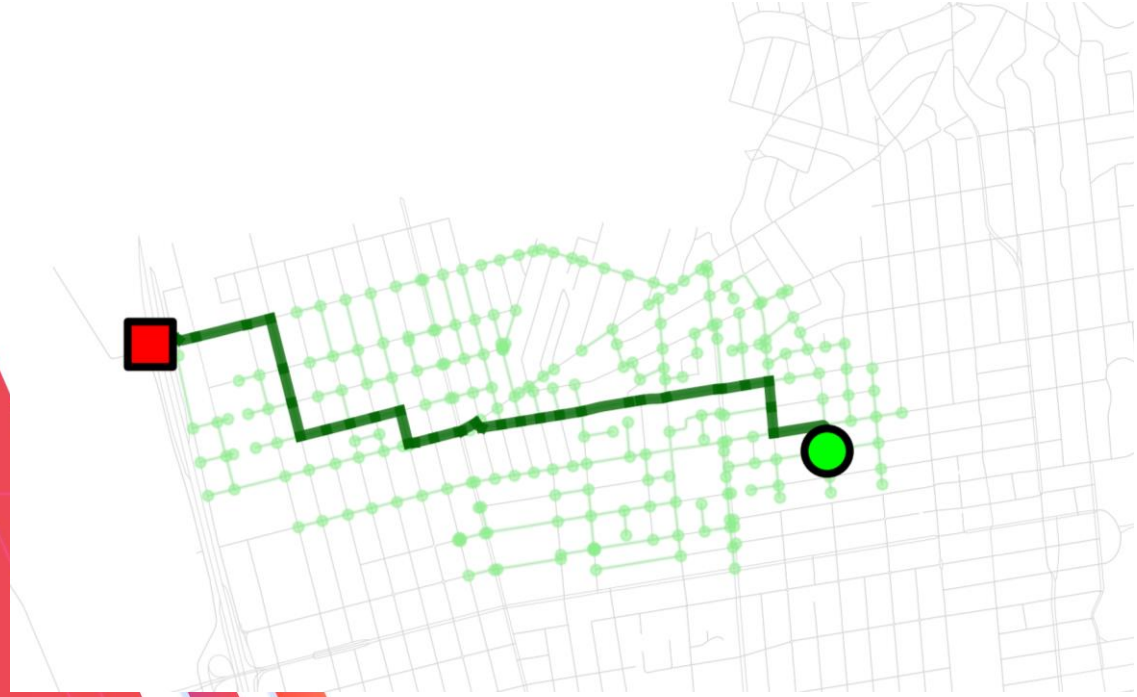
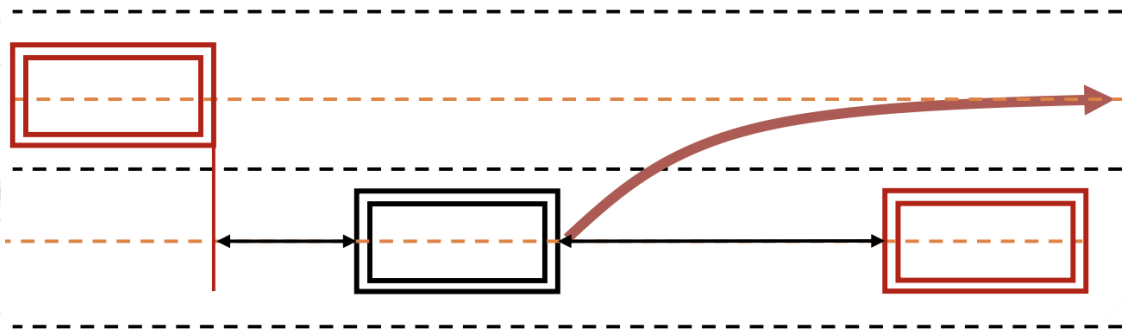
LiDAR Localization is also adopted in Apollo to provide precise position of the vehicle on the HD Map.





Localization in Apollo: LiDAR Localization



Introduction to Planning

Planning Module focuses on two different scales of path planning.
First is **route navigation**, which generates the route from A to B on a HD Map.
Then, subtle decisions will be made in **trajectory planning**, which focuses on avoiding the obstacles on the planned path, which lane to drive, etc.